AgentOS - Project Specification

1. Project Overview

AgentOS is an AI-powered private research assistant that combines Retrieval-Augmented Generation (RAG), OCR, conversation memory, and hybrid web search. Users can upload PDFs, DOCX, TXT, CSV files, and images, then ask natural-language questions to retrieve accurate information.

2. Application Scenarios

• Research assistance
• Personal knowledge management
• Enterprise document search
• Educational learning
• OCR-based image understanding

3. Agent Architecture

User → Streamlit UI → RAG Engine → ChromaDB → Groq Llama 3.3
Images are processed with EasyOCR before indexing. Tavily Web Search is used for current information.

4. Core Capabilities

• Multi-document RAG
• OCR for images
• Hybrid search
• Conversation memory
• Document summarization
• Source attribution

5. Technology Stack

Python, Streamlit, ChromaDB, Sentence Transformers, Groq Llama 3.3, EasyOCR, Tavily Search API.

6. Local Deployment Plan

Clone the repository, install requirements, configure API keys in a .env file, and launch with Streamlit. The application supports execution on AMD Radeon Cloud.

7. AMD Radeon GPU / ROCm Adaptation

AgentOS was deployed and validated on AMD Radeon Cloud. The deployment demonstrates compatibility with the AMD software ecosystem and cloud GPU environment.

8. Future Enhancements

Voice assistant support, enterprise connectors, multi-user authentication, local GPU inference, and advanced multimodal models.

9. Team

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